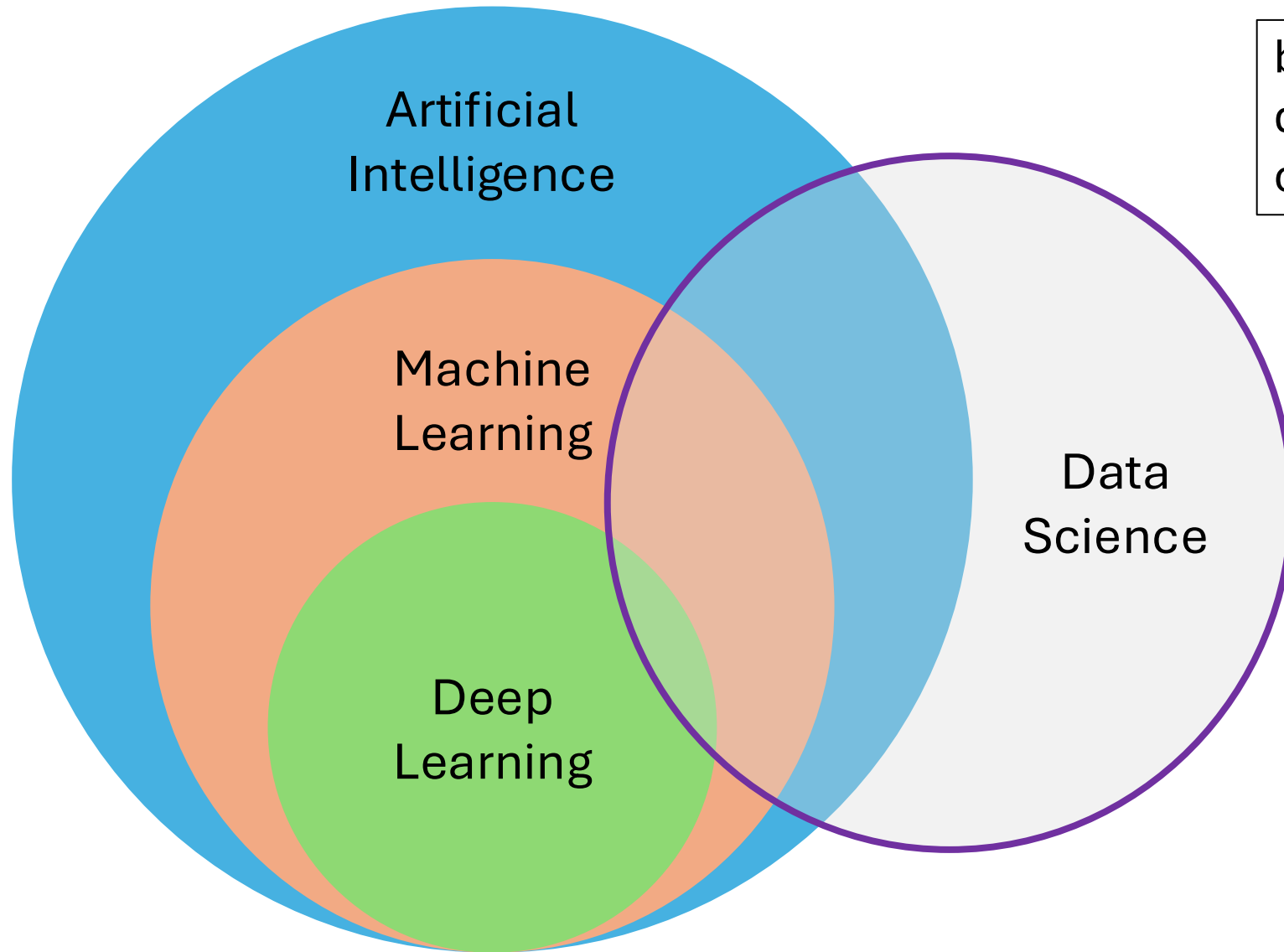


Introduction

Data Science

Course Schedule

1. Exploratory Data Analysis
2. Clustering
3. Probabilistic Models
4. Statistical Inference
5. Machine Learning
6. Feature Engineering
7. Model Evaluation
8. Deep Learning
9. Time Series
10. Causality



blend of diverse components from different domains (statistics, optimization, computer science, ...)

Deep Learning: special kind of ML methods using *deep* neural networks (e.g., CNNs, transformers)

Data Science:
extract knowledge from data (by means of ML, among other things)

Typical Data Science Workflow

Business Understanding

→ Data Collection

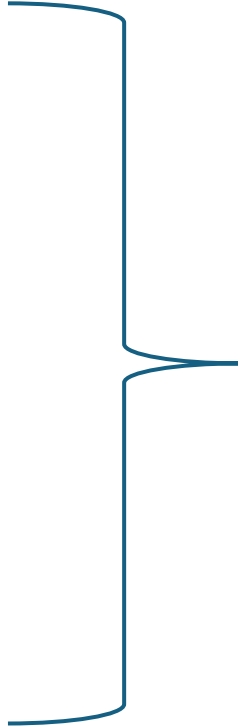
→ Data Exploration

→ Data Cleaning & Preparation

→ Modeling

→ Evaluation

→ Deployment & Monitoring



iterative process

Main Areas of Artificial Intelligence (AI)

tabular data

	ID	MSBSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LowShade	LandContour	Utilities		PoolArea	PoolQC	Fence	MiscFeature	MisVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice
1450	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	NaN	NaN	0	2	2008	WD	Normal	208500
	2	20	RL	89.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	NaN	NaN	0	5	2007	WD	Normal	181500
	3	70	RL	68.0	11250	Pave	NaN	TRI	Lvl	AllPub	...	0	NaH	NaN	NaN	0	9	2008	WD	Normal	223500
	4	70	RL	60.0	9550	Pave	NaN	TRI	Lvl	AllPub	...	0	NaH	NaN	NaN	0	2	2006	WD	Abnorml	146000
	5	60	RL	84.0	14260	Pave	NaN	TRI	Lvl	AllPub	...	0	NaH	NaN	NaN	0	12	2008	WD	Normal	250000
1455	1456	60	RL	62.0	7917	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	NaN	NaN	0	8	2007	WD	Normal	175000
1456	1457	70	RL	85.0	13175	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	MnPrv	NaN	0	2	2010	WD	Normal	210000
1457	1458	70	RL	66.0	9042	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	GdPrv	Shed	2500	5	2010	WD	Normal	260500
1458	1459	70	RL	69.3	9717	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	NaN	NaN	0	2	2010	WD	Normal	142100
1459	1460	20	RL	75.0	9937	Pave	NaN	Reg	Lvl	AllPub	...	0	NaH	NaN	NaN	0	6	2008	WD	Normal	147500
[1460 rows x 21 columns]																					

language models

The Lord of the Rings

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from *Lara of the Rings*)

*This article is about the book. For other uses, see **The Lord of the Rings** (disambiguation). 'War of the Ring' redirects here. For other uses, see **War of the Rings** (disambiguation).*

The Lord of the Rings is an epic^[1] *high fantasy* novel^[2] by the English author and scholar J. R. R. Tolkien. Set in Middle-earth, it is regarded as a sequel to Tolkien's 1937 children's book *The Hobbit*, but eventually developed into a much larger work. Written in stages between 1937 and 1949, *The Lord of the Rings* is one of the best-selling books ever written, with over 150 million copies sold.^[3]

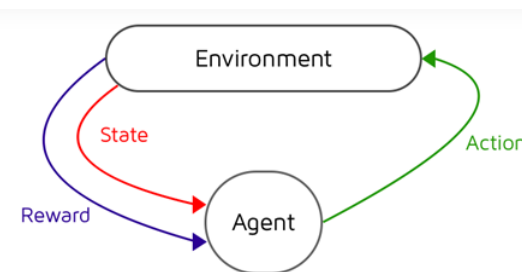
The title refers to the story's main antagonist **3 Rings of Power**, the Dark Lord who in an earlier age created the **One Ring** to rule the other **Rings of Sauron** given to Men, Elves, and Ents. In his campaign to conquer all of Middle-earth, from homely beginnings in **Shire**, a hobbit land reminiscent of the English countryside, the story ranges across Middle-earth, following the quest to destroy the **One Ring**, seen mainly through the eyes of the hobbits **Frodo**, **Sam**, **Merry**, and **Pippin**. **Arwen** Frodo's love is the Wizard **Gandalf**, **Men** **Arwen** and **Boromir**, the **Elves**, **gulls**, and the **Dwarf King**, who unite in order to free the **Free Peoples** of Middle-earth against Sauron's armies and give Frodo a chance to destroy the **One Ring** in the fire of **Mount Doom**.

Although often mistakenly called a trilogy, the work is intended by Tolkien to be one volume in a two-volume set along with *The Silmarillion*.^{[4][5]} For economic reasons, *The Lord of the Rings* was first published over the course of a year from 29 July 1954 to 20 October 1955 in three volumes rather than one^[4] under the titles *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*. The *Silmarillion* appeared only after the author's death, the story was divided internally into six books, two per volume, with several appendices of background material.^[4] These three volumes were later published as a boxed set, and eventually finally as a single volume, following the author's original intent.

computer vision



control

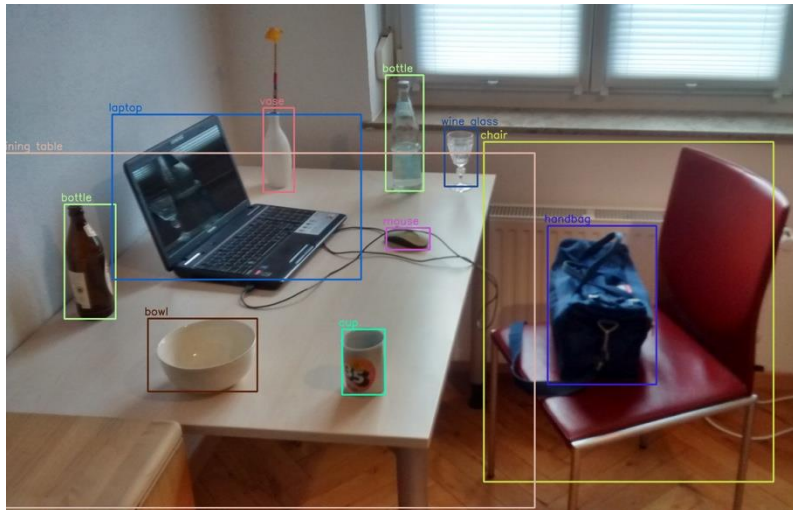


all empowered by one key component:
learning from data → Machine Learning (ML)

When to Use ML (Learning from Data)

automation

too complex for rules



object recognition, chatbot, ...

uncertainty

too complex for humans



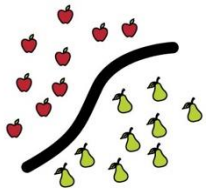
protein structure predictions, demand forecasting, ...

MACHINE LEARNING

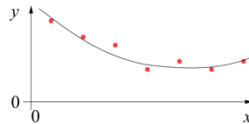
training target available
(labeled or past data)

SUPERVISED

CLASSIFICATION



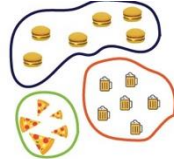
REGRESSION



data not labeled
in any way

UNSUPERVISED

CLUSTERING

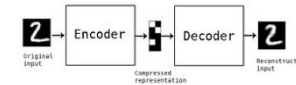
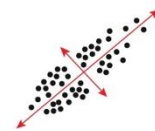


SELF-SUPERVISED

ASSOCIATION



DIMENSIONALITY REDUCTION

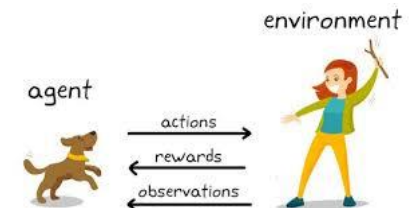


no supervision, but goal-based
interaction with environment

REINFORCEMENT LEARNING

LEARN STATE OR ACTION VALUES

LEARN POLICY DIRECTLY



learning by teacher
(high-dimensional curve fitting)

learning by observation
(pattern recognition)

learning by trial-and-error
(sequential decision making)



Used Programming Language: Python

- good compromise between rapid prototyping and production
- vast ecosystem
- very popular for data processing and ML: scientific Python stack

Scientific Python Stack



Programming Environments

locally, best use

- a virtual environment to flexibly install packages (e.g., [venv](#))
- an IDE of your choice (e.g., [VS Code](#))
- both plain Python files or [Jupyter notebooks](#) are fine

but cloud-based environments also fine (e.g., [Google Colab](#))

Literature

- [An Introduction to Statistical Learning](#)
- [The Elements of Statistical Learning](#)
- [Pattern Recognition and Machine Learning](#)
- [Machine Learning](#)

- [Deep Learning](#)
- [Understanding Deep Learning](#)